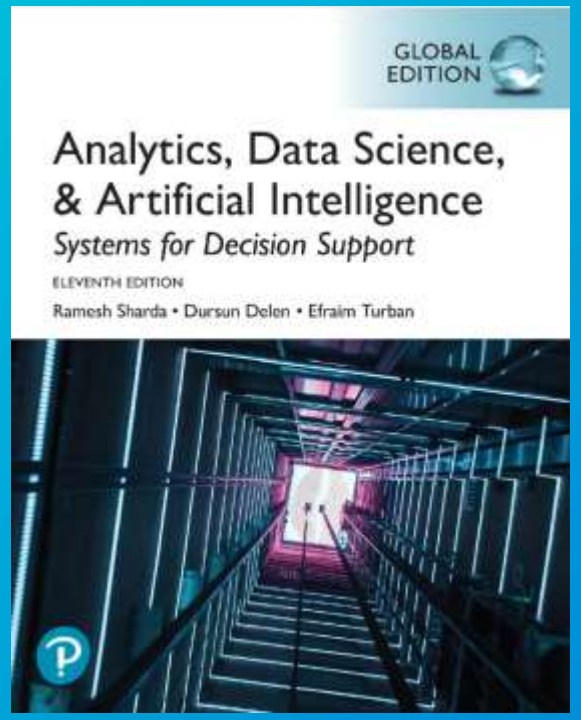
MANAGEMENT SUPPORT SYSTEMS
(415 MIS-3)
CHAPTER – 4



Chapter 8 Prescriptive Analytics: Optimization and Simulation

Analytics, Data Science, & Artificial Intelligence Systems For Decision Support,11ed, Global Edition

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8.2 Model-Based Decision Making

- Prescriptive Analytics Model Examples
- Identification of the Problem and Environmental Analysis
- Model Categories
- Current Trends in Modeling

Prescriptive Analytics Model Examples

- **Modeling** is a key element for prescriptive analytics.
- Depending on the problem we are addressing, there are many classes of models, and there are often many specialized techniques for solving each one.
- Prescriptive analytics involves the application of mathematical models, sometimes the term data science is more commonly associated with the application of such mathematical models.

Identification of the Problem and Environmental Analysis

- It is important to analyze the scope of the domain and the forces and dynamics of the environment when making a decision.
- A decision maker needs to identify the organizational culture and the corporate decision-making processes.
- **Environmental scanning** and analysis is the monitoring, scanning, and interpretation of collected information.
- Variable Identification is critical, as are the relationships among the variables.
- **Influence diagrams** can facilitate the identification process.
- A **cognitive map**, can help a decision maker develop a better understanding of a problem, and variable interactions.

Identification of the Problem and Environmental Analysis

- **Predictive analytics** (forecasting) is essential for constructing and manipulating models because the results of an implemented decision occur in the future.
- There is no point in running a what-if (sensitivity) analysis on the past because decisions made then have no impact on the future.
- Online commerce and communication has created an immense need for forecasting and an abundance of available information for performing it.
- These activities occur quickly, yet information about such purchases is gathered and should be analyzed to produce forecasts.
- Forecasting models use product life-cycle needs and information about the marketplace to analyze the situation.

Model Categories

- The following table classifies decision models into seven groups and lists several representative techniques for each category.
- Each technique can be applied to a static or a dynamic model, which can be constructed under assumed environments of certainty, uncertainty, or risk.

TABLE 8.1 Categories of Models

Category	Process and Objective	Representative Techniques
Optimization of problems with few alternatives	Find the best solution from a small number of alternatives	Decision tables, decision trees, analytic hierarchy process
Optimization via algorithm	Find the best solution from a large number of alternatives, using a step-by-step improvement process	Linear and other mathematical programming models, network models
Optimization via an analytic formula	Find the best solution in one step, using a formula	Some inventory models
Simulation	Find a good enough solution or the best among the alternatives checked, using experimentation	Several types of simulation
Heuristics	Find a good enough solution, using rules	Heuristic programming, expert systems
Predictive models	Predict the future for a given scenario	Forecasting models, Markov analysis
Other models	Solve a what-if case, using a formula	Financial modeling, waiting lines

Activate Windows

Model Categories (cont.)

- To expedite model construction, we can use special decision analysis systems that have modeling languages and capabilities embedded in them.
- These include spreadsheets, data mining systems, online analytic processing (OLAP) systems, and modeling languages.
- **Model Management:** Models must be managed to maintain integrity, and applicability. This is done with the aid of model-based management systems, which are analogous to database management systems (DBMS).
- **Knowledge-based Modeling DSS**: use quantitative models, whereas expert systems use qualitative, knowledge-based models in their applications.

Current Trends in Modeling

- One trend in modeling involves the development of model libraries and solution technique libraries.
- There is a clear trend toward developing and using cloud-based tools and software to run software to perform modeling, optimization, simulation, etc.
- With management models, the amount of data and model sizes is large, necessitating data warehouses and parallel computing for solutions.
- There is a trend toward making analytics models transparent to decision makers, and using influence diagrams (a model of a model to help in analysis).
- Many decision makers accustomed to slicing and dicing data cubes are now using OLAP systems that access data warehouses.

8.3 Structure of Mathematical Models for Decision Support

- Components of Decision Support Mathematical Models
- The Structure of Mathematical Models

Components of Decision Support Mathematical Models

- Quantitative models are made up of **four** basic components: result variables, decision variables, uncontrollable variables, and intermediate result variables.
- Mathematical relationships link these components together.
- In non-quantitative models, the relationships are symbolic or qualitative. The results of decisions are determined based on decisions made, uncontrollable variable, and relationships among variables.
- The modeling process involves identifying the variables and relationships. Solving a model determines the values of these and the result variable(s).

Components of Decision Support Mathematical Models

- **Result/Outcome/Output Variables:** reflect level of effectiveness of a system.
- **Decision Variables:** Decision variables describe alternative courses of action. The decision maker controls the decision variables.
- **Uncontrollable Variables/Parameters:** fixed/varying factors that affect the result variables but are not under the decision maker control.
- **Intermediate Result Variables:** reflect intermediate outcomes in models.

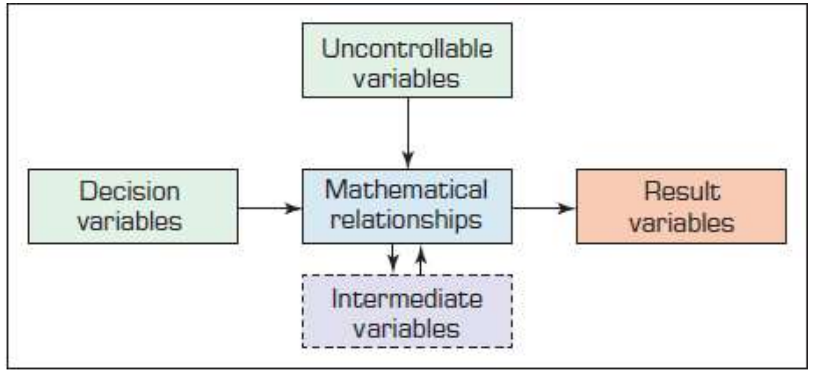


FIGURE 8.1 The General Structure of a Quantitative Model.

The Structure of Mathematical Models

- Components of a quantitative model are linked by mathematical expressions.
- A very simple financial model is: $P = R - C$
- Another financial model is the simple present-value cash flow model.
- It is possible to determine the present value of a payment of $100,000 to be made 5 years from today, at a 10% interest rate, as follows:

$$P = 100{,}000/(1 + 0.1)^5 = 62{,}092$$

8.4 Certainty, Uncertainty, and Risk

- Decision Making under Certainty
- Decision Making under Uncertainty
- Decision Making under Risk (Risk Analysis)

Decision Making under Certainty

- In decision making under certainty, complete knowledge is available so decision maker know the outcome of each course of action.
- This is done with structured problems and short time horizons (up to 1 year).
- Outcomes are not 100% known, but this assumption simplifies the model.
- The decision maker is viewed as a perfect predictor of the future because it is assumed that there is only one outcome for each alternative.
- Certainty models are easy to develop and solve, and yield optimal solutions.
- Financial models are constructed under assumed certainty.

Decision Making under Uncertainty

- Decision maker considers situation where several outcomes are possible for each course of action.
- In contrast to the risk situation, the decision maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes.
- Modeling such situations involves assessment of the decision maker's attitude toward risk.
- Instead of dealing with uncertainty, manager's sometimes attempt to obtain more information so that the problem can be treated under certainty.
- If more information is not available, the problem must be treated under a condition of uncertainty, which is less definitive than the other categories.

Decision Making under Risk (Risk Analysis)

- **Risk analysis** is a decision-making method that analyzes risk associated with alternatives, each with a given probability of occurrence.
- The probabilities that the given outcomes will occur are assumed to be known or can be estimated. Under these assumptions, the decision maker can assess the degree of risk associated with each alternative (calculated risk).
- Risk analysis can be performed by calculating the expected value of each alternative and selecting the one with the best expected value.

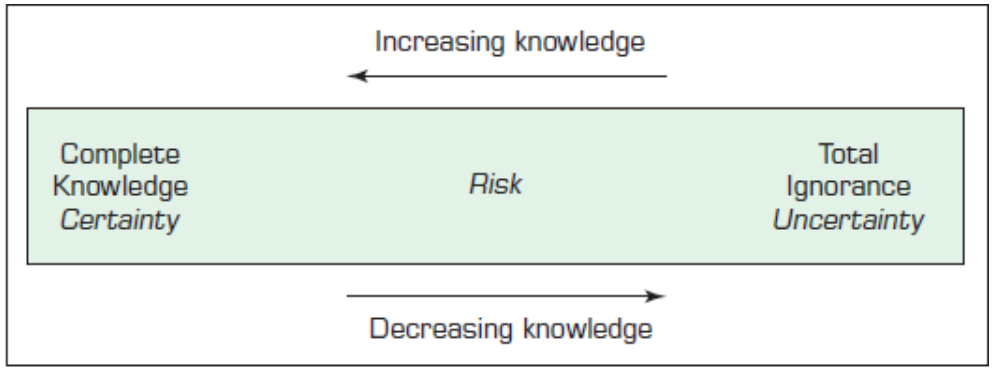


 FIGURE 8.2 The Zones of Decision Making.

8.5 Decision Modeling with Spreadsheets

- Application Case 8.4: Pennsylvania Adoption Exchange Uses Spreadsheet Model to Better Match Children with Families
- Application Case 8.5: Metro Meals on Wheels Treasure Valley Uses Excel to Find Optimal Delivery Routes
- Modelling with Spreadsheets

Decision Modeling with Spreadsheets

- Models can be developed and implemented in a variety of programming languages and systems. We focus primarily on spreadsheets, modeling languages, and transparent data analysis tools.
- Spreadsheet packages were recognized as easy-to-use software due to their strength and flexibility.
- Spreadsheet packages were used in the development of a wide range of applications in business, mathematics, and science.
- As spreadsheet packages evolved, add-ins were developed for structuring and solving specific model classes.
- Among the add-in packages, many were developed for DSS development.

Application Case 8.4: Pennsylvania Adoption Exchange Uses Spreadsheet Model to Better Match Children with Families

- The Pennsylvania Adoption Exchange (PAE) was established to help county and nonprofit agencies find prospective families for orphan children who had not been adopted due to age or special needs.
- The PAE started collecting information about the orphans and families through online surveys that include a new set of questions. The PAE and consultants created a spreadsheet matching tool. In this model, caseworkers can specify the weight of the attributes for selecting a family for a child.
- Using this tool, the matching committee can compare a child and family on each attribute, thus making a more accurate match decision between a family and a child.
- **Results/Benefits:** Since use of the new spreadsheet, the percentage of children getting a permanent home has increased.

Application Case 8.5: Metro Meals on Wheels Treasure Valley Uses Excel to Find Optimal Delivery Routes

- Meals on Wheels Association of America is a not-for-profit organization that delivers approximately one million meals to homes of older people in need.
- Metro Meals on Wheels wanted a routing tool that could generate routing solutions for both one-way and round-trip directions for delivering meals.
- To solve the routing problem, a spreadsheet-based tool was developed.
- Excel's Visual Basic was used to access a developer's networking map application programming interface (API) called MapQuest.
- This API was used to create a travel matrix that calculated time and distance needed for delivery of the meal.
- **Results/Benefits**: Metro Meals saved $5,800 in 2015. The tool reduced the time spent on route planning for meal deliveries. Also, It increased volunteer satisfaction.

Modeling with Spreadsheets

- **Goal seeking** is an important spreadsheet feature. It is performed by indicating a target cell, its desired value, and a changing cell.

  - A change in the interest rate is immediately reflected in the monthly payment.
  - If we require a specific monthly payment, we can use goal seeking to determine an appropriate interest rate or loan amount.
  - The results can be observed and analyzed immediately.
  - The **static model** indicates a single month's performance, which is replicated.

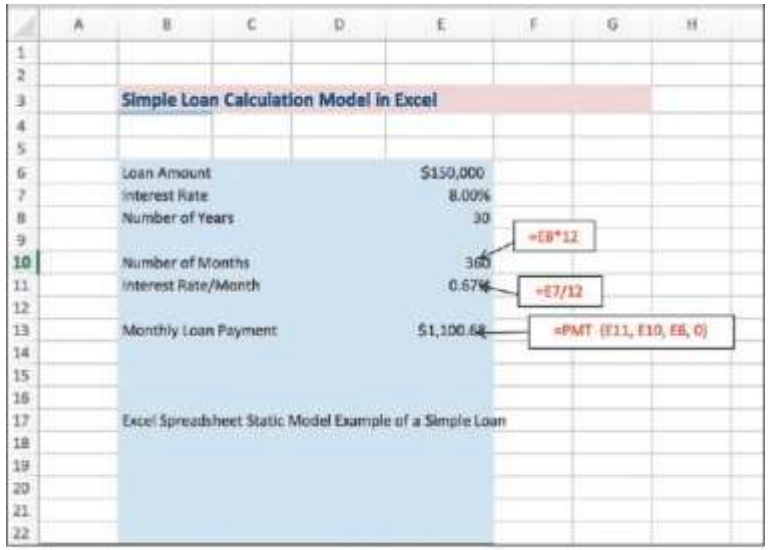


FIGURE 8.3 Excel Spreadsheet Static Model Example of a Simple Loan Calculation of Monthly Payments.

Modeling with Spreadsheets (cont.)

- A **dynamic model**, in contrast, represents behavior over time.

- Risk analysis can be incorporated into spreadsheets by using built-in random-number generators to develop simulation models.

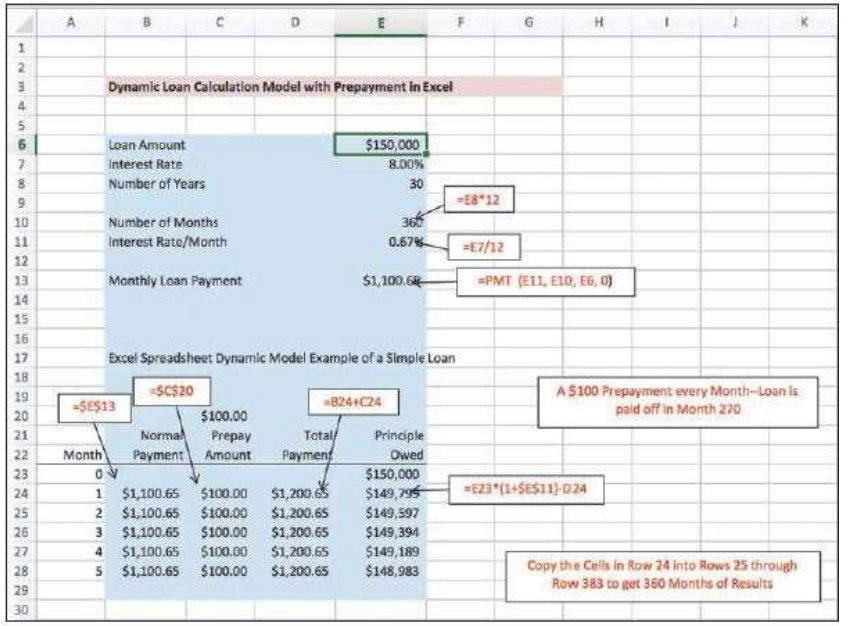


FIGURE 8.4 Excel Spreadsheet Dynamic Model Example of a Simple Loan Calculation of Monthly Payments and the Effects of Prepayment.

8.6 Mathematical Programming Optimization

- Overview
- Linear Programming Model
- Implementation

Overview

- Mathematical programming are tools that helps decision makers allocate scarce resources among competing activities to optimize a measurable goal.
- In linear programming (LP), one of the tools, all variable relationships are linear. Applications include supply chain management, product decisions, etc.

LP allocation problems usually display the following characteristics:

- Limited quantity of resources, most are used in product/service production
- Two or more ways resources can be used. Each is called a solution/program.
- Each activity where resources are used, yields a return in terms of stated goal

Linear Programming Model

- **The LP allocation model is based on the following economic assumptions:**
  - Returns from allocations are independent and measured by a common unit
  - The total return is the sum of the returns yielded by the different activities.
  - All data are known with certainty, and resources are used economically
- Allocation problems have a large number of possible solutions. Depending on the assumptions, the number of solutions can be either infinite or finite.
- Different solutions yield different rewards. The solution with highest degree of goal attainment is called optimal solution, and found by a special algorithm

Linear Programming Model (cont.)

- **Every LP model is composed of:**
  - Decision variables: Unknown values that are being searched for
  - Objective Function: A linear mathematical function that relates the decision variables to the goal, measures goal attainment, and is to be optimized
  - Coefficients: indicate contribution to objective of one unit of a decision variable
  - Constraint: Linear (in)equalities that limit resources
  - Capacities: Describe upper and lower limits on the constraints and variables
  - Input/output & Coefficients: Indicate resource utilization for a decision variable

Implementation

- Implement the model in "standard form", where constraints are written with decision variables on the left and a number on the right.
- Alternatively, use spreadsheet to calculate the model in a less rigid manner.
- LP models can be specified directly in a number of user-friendly modeling systems. Models are specified in the same way they are defined algebraically.
- **Optimization models can be solved by mathematical programming methods:**
  - Assignment & Network models for planning and scheduling
  - Dynamic, Goal, Linear, Nonlinear and Integer programming
  - Investment & Replacement
  - Simple inventory models & Transportation

8.6 Mathematical Programming Optimization

- Modeling in LP: An Example

Modeling in LP: An Example

- **MBI Corporation**, which manufactures special-purpose computers, needs to **make a decision**: How many computers should it produce next month at the Boston plant?
- MBI is considering two types of computers:
  - CC-7, which requires 300 days of labor and $10,000 in materials, &
  - CC-8, which requires 500 days of labor and $15,000 in materials.
- The profit contribution of each CC-7 is $8,000, and of each CC-8 is $12,000.
- The plant has a capacity of 200,000 working days per month, and the material budget is $8 million per month.
- Marketing requires that at least 100 units of the CC-7 and at least 200 units of the CC-8 be produced each month.
- **Problem:** Maximize the company's profits by determining how many units of the CC-7 and how many units of the CC-8 should be produced each month.

Modeling in LP: An Example (cont.)

- The **problem** is to find the values of the decision variables *X*1, *X*2, such that the value of the result variable *Z* is maximized, subject to a set of linear constraints that express the technology, market conditions, and other uncontrollable variables.
- The mathematical relationships are all linear equations and inequalities.

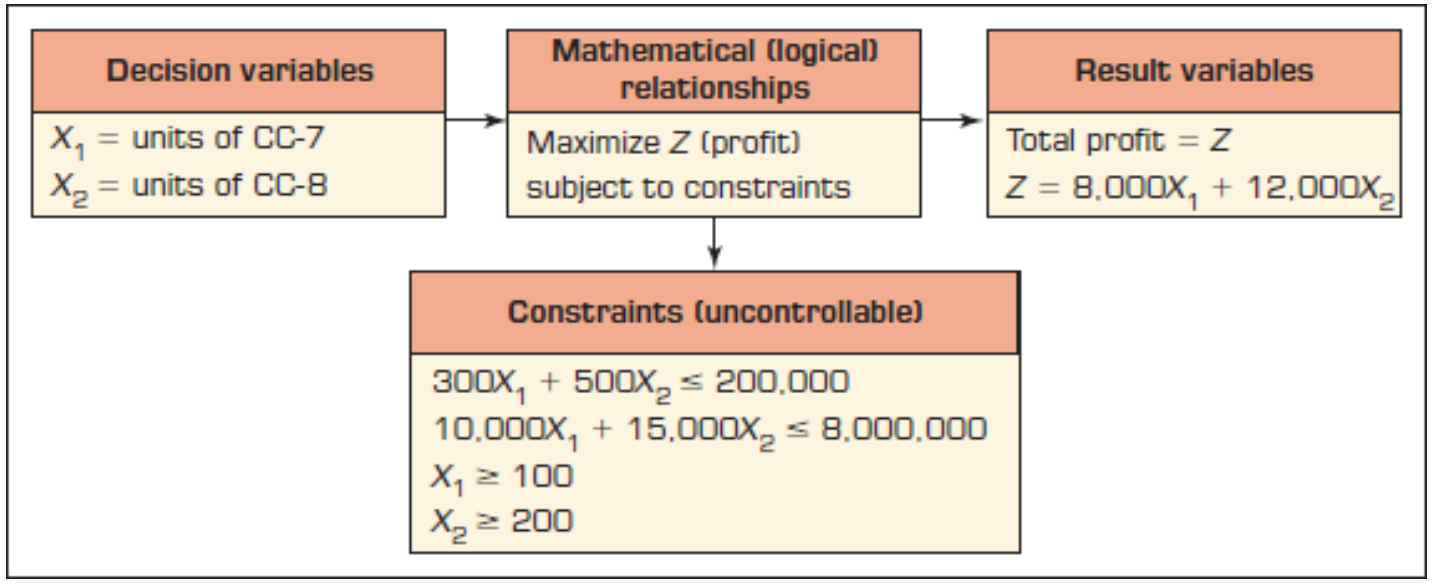


FIGURE 8.5 Mathematical Model of a Product-Mix Example.

Modeling in LP: An Example (cont.)

- Theoretically, any allocation problem of this type has an infinite number of possible solutions.
- Using special mathematical procedures, the LP approach applies a unique computerized search procedure that finds the best (optimal) solution/s (ex. maximizes total profit) in a matter of seconds.
- Excel 'add-in Solver' is used to obtain an optimal (best) solution to this problem.

    1. Activate 'Solver' Under the Data tab and on the Analysis ribbon.

        If it is not there, you should be able to enable it by going to File -> Excel's Options Menu and selecting Add-ins.

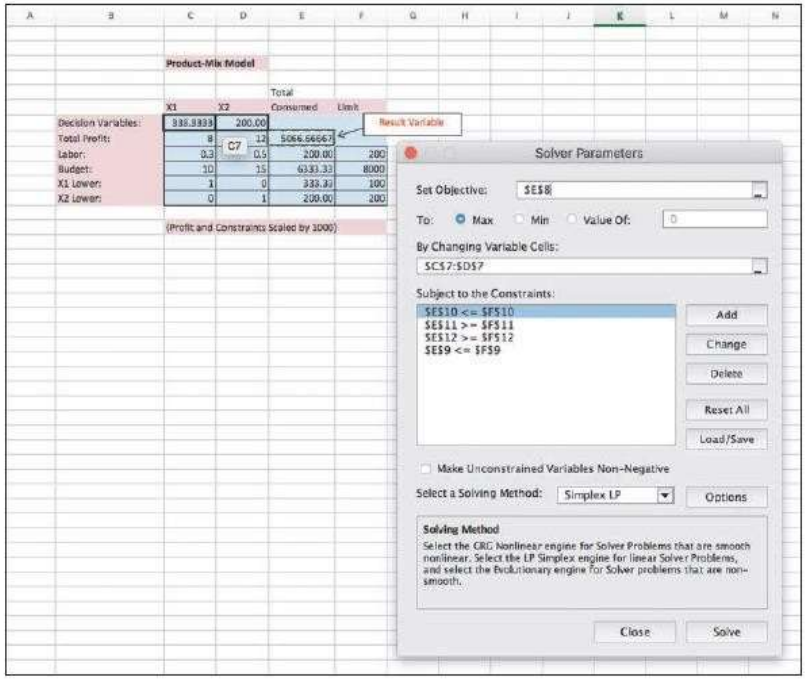


FIGURE 8.6 Excel Solver Solution to the Product-Mix Example.

Modeling in LP: An Example (cont.)

2. Enter these data directly into an Excel spreadsheet.
3. Identify the goal (by setting Target Cell equal to **Max**).
4. Identify decision variables (by setting By Changing Cells).
5. Identify constraints on labor capacity, budget, and the desired minimum production of the two products *X*1 and *X*2.
6. Clicking on the Solver Add-in opens a dialog box,
   - Specify the cells or ranges that define the objective function cell, decision/changing variables (cells), and the constraints.
   - Select the solution method (usually Simplex LP)
   - solve the problem.
6. Next, we select all three reports—Answer, Sensitivity, and Limits (optimal solution of *X*1 = 333.33, *X*2 = 200, Profit = $5,066,667)

8.7 Multiple Goals, Sensitivity Analysis, What-If Analysis, and Goal Seeking

- Multiple Goals
- Sensitivity Analysis
- Types of Sensitivity Analysis
- What-If Analysis
- What-If Analysis Example
- Goal Seeking
- Goal Seeking Example

Multiple Goals

- Today's management systems are complex, and managers want to attain simultaneous goals, some of which may conflict.
- Transform a multiple-goal problem into a single-measure-of effectiveness problem before comparing the effects of the solutions.
- **Certain difficulties may arise when analyzing multiple goals:**
  - It is difficult to obtain the organization's goals explicitly
  - Goals are viewed differently at various levels of the organization
  - Goals and their importance change in response to the organization
  - Complex problems are solved by decision makers with different agendas

Sensitivity Analysis

- Sensitivity analysis assesses impact of input data changes on proposed solution.

- **Sensitivity analysis allows for:**
  - Adaptation to conditions of different decision-making situations
  - Provides a better understanding of the model
  - Permits the input of data to increase model confidence.

Sensitivity Analysis (cont.)

Sensitivity analysis tests relationships such as the following:

- Impact of parameter change, and decision variables on outcome variable(s)
- The effect of uncertainty in estimating external variables
- The effect of different dependent interactions among variables
- The robustness of decisions under changing conditions

Sensitivity analyses are used for:

- Revising models to eliminate too-large sensitivities
- Detailing variables and obtaining estimates of sensitive external variables
- Altering a real-world system to reduce actual sensitivities

Types of Sensitivity Analysis

- The two types of sensitivity analyses are automatic and trial and error.

Automatic Sensitivity Analysis:

- This is performed in standard quantitative model implementations such as LP.
- It is usually limited to one change at a time, and only for certain variables.
- It is powerful because of its ability to establish ranges and limits very fast

Trial-and-error Sensitivity Analysis:

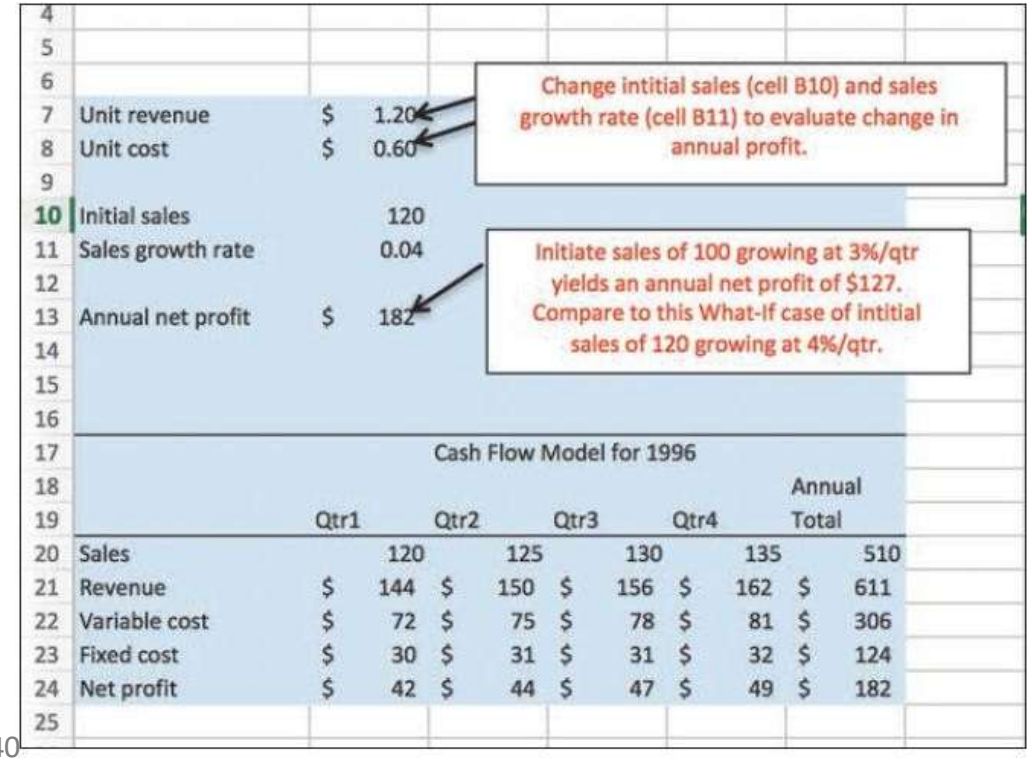
- Impact of changes in variable(s) is determined by trial-and-error approach
- When changes are repeated, better and better solutions may be discovered.
- Such experimentation has **two** approaches: what-if analysis and goal seeking.

What-If Analysis

- **What-if analysis** is structured as: *What will happen to the solution if an input variable, assumption, or parameter value is changed?*
  - Total inventory cost if the carrying inventories cost increases by 10%?
  - Market share if the advertising budget increases by 5%?
- With the appropriate user interface, managers can ask a computer model these types of questions and get immediate answers.
- Performs multiple cases and change the percentage, or other data as needed.
- What-if analysis is common in many decision systems. Users are given the opportunity to change their answers to the system's questions, and a revised recommendation is found.

What-If Analysis Example

A what-if query for a cash flow problem: the user changes the cells containing the initial sales (from 100 to 120) and the sales growth rate (from 3% to 4% per quarter), the program immediately re-computes the value of the annual net profit cell (from $127 to $182) .



7	Unit revenue	$	1.20
8	Unit cost	$	0.60
10	Initial sales		120
11	Sales growth rate		0.04
13	Annual net profit	$	182

Change intitial sales (cell B10) and sales growth rate (cell B11) to evaluate change in annual profit.

Initiate sales of 100 growing at 3%/qtr yields an annual net profit of $127. Compare to this What-If case of intitial sales of 120 growing at 4%/qtr.

Cash Flow Model for 1996

		Qtr1	Qtr2	Qtr3	Qtr4	Annual Total
20	Sales	120	125	130	135	510
21	Revenue	$ 144	$ 150	$ 156	$ 162	$ 611
22	Variable cost	$ 72	$ 75	$ 78	$ 81	$ 306
23	Fixed cost	$ 30	$ 31	$ 31	$ 32	$ 124
24	Net profit	$ 42	$ 44	$ 47	$ 49	$ 182

Goal Seeking

- **Goal seeking** calculates the values of the inputs necessary to achieve a desired level of an output (goal). The following are some examples:

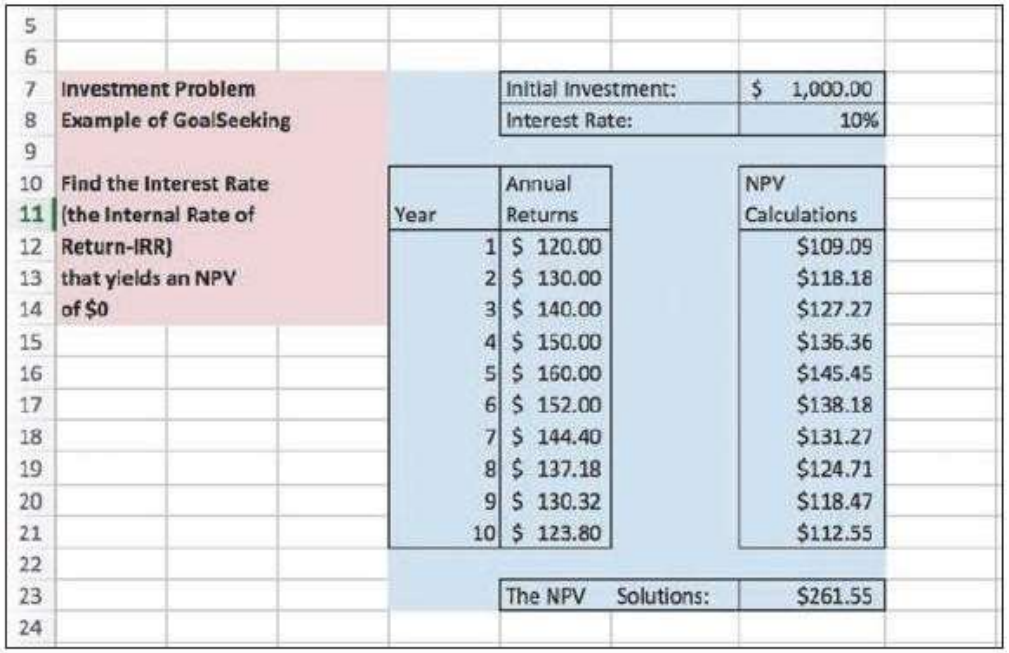
  - Annual R&D budget is needed for an annual growth rate of 15% by 2018?
  - How many nurses needed to reduce the average waiting time of a patient in the emergency room to less than 10 minutes?

- **Computing A Break-even Point By Using Goal Seeking:**

  - Some modeling software packages can directly compute break-even points, which is an important application of goal seeking.
  - Determining the value of the decision variables that generate zero profit.
  - Sensitivity analysis as the prewritten routines present a limited opportunity for asking what-if questions.

Goal Seeking

- In a financial planning model, the internal rate of return (IRR) is the interest rate that produces a net present value (NPV) of zero.
- Given a stream of annual returns in Column E, we can compute the NPV of planned investment through goal-seeking.
- An NPV equal to zero determines the IRR of this cash flow, including the investment. We set the NPV cell to 0 by changing the interest rate cell.
- **The answer is 38.77059%.**



5					
6					
7	**Investment Problem**		Initial Investment:	$ 1,000.00	
8	**Example of GoalSeeking**		Interest Rate:	10%	
9					
10	**Find the Interest Rate**	Year	Annual Returns	NPV Calculations	
11	**(the Internal Rate of**				
12	**Return-IRR)**	1	$ 120.00	$109.09	
13	**that yields an NPV**	2	$ 130.00	$118.18	
14	**of $0**	3	$ 140.00	$127.27	
15		4	$ 150.00	$136.36	
16		5	$ 160.00	$145.45	
17		6	$ 152.00	$138.18	
18		7	$ 144.40	$131.27	
19		8	$ 137.18	$124.71	
20		9	$ 130.32	$118.47	
21		10	$ 123.80	$112.55	
22					
23			The NPV Solutions:	$261.55	
24					

8.8 Decision Analysis with Decision Tables & Decision Trees

- Decision Tables
- Decision Tables Example
- Decision Trees

Decision Tables

- Decision tables organize information in systematic, tabular form for analysis.
- **Treating Uncertainty**: Several methods are available for handling uncertainty.
  - Optimistic approach: assumes and selects best outcomes for alternatives
  - Pessimistic approach: assumes worst outcome for alternatives; selects the best
  - Another approach simply assumes that all states of nature are equally possible.
- When possible, analysts should attempt to gather information to treat the problem under assumed certainty.
- **Treating Risk:** The most common method for solving this risk analysis problem is to select the alternative with the greatest expected value.

Decision Tables Example

- An investor estimates: solid growth (50%), stagnation (30%), and inflation (20%)
- Expected value is computed by multiplying result probabilities and adding them

TABLE 8.3 Investment Problem Decision Table Model

	State of Nature (Uncontrollable Variables)		
Alternative	Solid Growth (%)	Stagnation (%)	Inflation (%)
Bonds	12.0	6.0	3.0

Bond investment yields an expected return of 12(0.5) + 6(0.3) + 3(0.2) = 8.4%

- Suppose a financial advisor presents a $1,000 investment with 0.9999 chance to double your money, and 0.0001 chance you'll lose $500,000.

0.9999 ($2,000 - $1,000) + .0001(-$500,000 - $1,000) = $999.90 - $50.10 = $949.80

- The potential loss could be catastrophic for any investor

Decision Trees

- **Decision trees** are alternative representations of a decision table; it shows a problem's relationship graphically and handles complex situations compactly.
- TreeAge Pro & PrecisionTree are systems that show decision trees in practice.
- You can apply mathematical programming to decision-making situations under risk. These include simulation, certainty factors, and fuzzy logic.
- A simplified investment case of multiple goals is shown in the table. The three goals are yield, safety, and liquidity. This situation is under assumed certainty.

TABLE 8.4 Multiple Goals

Alternative	Yield (%)	Safety	Liquidity
Bonds	8.4	High	High
Stocks	8.0	Low	High
CDs	6.5	Very high	High

8.9 Introduction to Simulation

- Major Characteristics of Simulation
- Advantages of Simulation
- Disadvantages of Simulation
- The Methodology of Simulation
- Simulation Types
- Monte Carlo Simulation
- Discrete Event Simulation

Major Characteristics of Simulation

- **Simulation** involves building a model of reality to the extent practical.
- Simulation models may suffer from fewer assumptions about the decision situation as compared to other prescriptive analytic models.
- Simulation is a technique for conducting experiments. Therefore, it involves testing specific values of the decision or uncontrollable variables in the model and observing the impact on the output variables.
- Simulation is used only when a problem is too complex to be treated using numerical optimization techniques.
- Complexity in this situation means either that the problem cannot be formulated for optimization, that the formulation is too large, that there are too many interactions among the variables.

Advantages of Simulation

Simulation is used in decision support modeling for the following reasons:

- The theory is straightforward, and model is built from manager's perspective.
- Time compression is attained quickly to give idea of policies' long-term effects.
- Descriptive rather than normative, allowing managers to ask what-if questions, and use a trial-and-error approach with less expense and risk.
- Requires intimate knowledge; model builder constantly interact with manager.
- Can handle a variety of problem types and higher-level managerial functions.
- Produces performance measures, and includes real complexities of problems.
- Can readily handle relatively unstructured problems.

Disadvantages of Simulation

- An optimal solution cannot be guaranteed, but relatively good ones are generally found.
- Simulation model construction can be a slow and costly process, although newer modeling systems are easier to use than ever.
- Solutions and inferences from a simulation study are usually not transferable to other problems because the model incorporates unique problem factors.
- Simulation is sometimes so easy to explain to managers that analytic methods are often overlooked.
- Simulation software sometimes requires special skills because of the complexity of the formal solution method.

The Methodology of Simulation

Simulation involves setting up a model of a real system through the steps:

- **Define the problem:** Examine problem, and specify need for simulation
- **Construct model:** Determine variables, relationships, and gather data.
- **Test and validate model:** Ensure model properly represents studied system.
- **Design experiment:** There are two conflicting objectives: accuracy and cost.
- **Conduct experiment:** can involve issues like number generation.
- **Evaluate results:** Statistical tools/sensitivity analyses used to interpret results.
- **Implement results:** Managerial involvement leads to implementation success.

The Methodology of Simulation

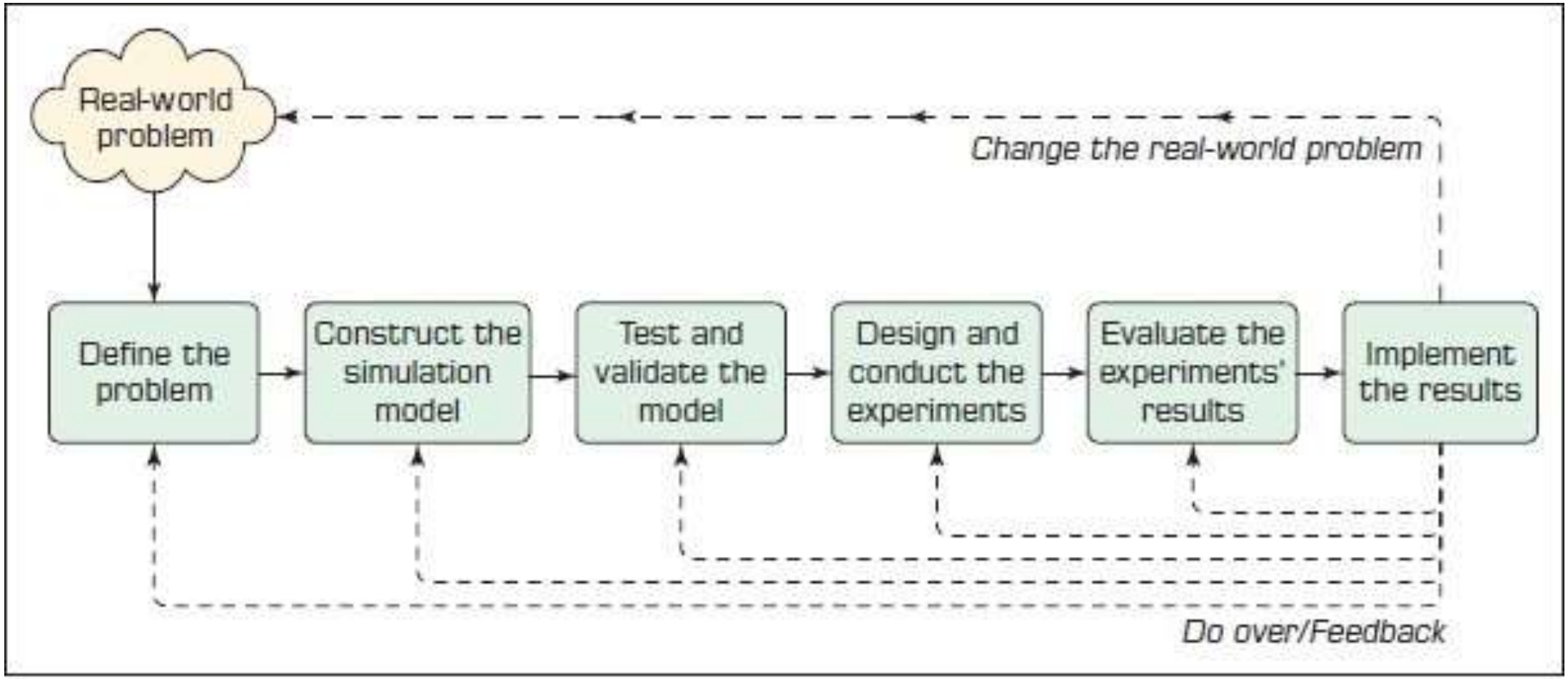


FIGURE 8.12 The Process of Simulation.

Simulation Types

- **Simulation model** consists of relationships that present the real-world operations. Simulation results depend on the set of parameters given as inputs.
- There are various simulation paradigms such as Monte Carlo simulation, discrete event, agent based, or system dynamics.
- The level of abstraction in a problem can determine simulation technique.
- Discrete events and agent based models are used for low levels of abstraction.
- They consider individual elements such as people in the simulation models, whereas systems dynamics is more appropriate for aggregate analysis.
- Here we introduce the major types of simulation: probabilistic simulation, time-dependent and time-independent simulation, and visual simulation.

Simulation Types

- **Probabilistic Simulation**: One or more independent variables are probabilistic. They follow probability distributions, which can be discrete or continuous:

  - Discrete involves situations with limited event numbers and finite values.
  - Continuous distributions are situations with unlimited numbers of possible events that follow density functions, such as the normal distribution.

- **Time-dependent Versus Time-independent Simulation**:

  - Time-independent refers to situations where time of event occurrence is unimportant. For example, we may know that the demand for a product is 3 units/day, but do not care when during the day the item is demanded.
  - However, in waiting-line problems applicable to e-commerce, it is important to know the precise time of arrival. This is a time-dependent situation.

Monte Carlo Simulation

- In business decision problems, we employ probabilistic simulations. The **Monte Carlo simulation** is commonly used.
- This method begins with building a model of the decision problem without having to consider the uncertainty of any variables.
- Then we recognize that certain variables are uncertain or follow an estimated probability distribution. This estimation is based on analysis of past data.
- Then we begin running sampling experiments. This consists of generating random values of uncertain parameters and then computing values of the variables that are impacted by such parameters or variables.
- We then analyze the behavior of these performance variables by examining their statistical distributions.

Discrete Event Simulation

- **Discrete event simulation** refers to building a model of a system where the interaction between different entities is studied.
- An example of this is modeling the customers arriving at various rates and the server serving at various rates, we can estimate the average system performance, waiting time, number of waiting customers, etc. Such systems are viewed as collections of customers, queues, and servers.
- There are thousands of documented applications of discrete event simulation models in engineering, business, and so on.
- Tools for building discrete event simulation models have been around for a long time, but these have evolved to take advantage of developments in graphical capabilities for building and understanding the results of such simulation models.

8.10 Visual Interactive Simulation

- Conventional Simulation Inadequacies
- Visual Interactive Simulation
- Visual Interactive Models and DSS

Conventional Simulation Inadequacies

- **Simulation** is a well-established, useful, descriptive, mathematics-based method for gaining insight into complex decision-making situations.
- Simulation does not usually allow decision makers to see how a solution to a complex problem evolves over time, nor interact with the simulation.
- Simulation reports statistical results at the end of a set of experiments. Decision makers are thus not an integral part of simulation development and experimentation, and their experience and judgment cannot be used directly.
- If the simulation results do not match the intuition or judgment of the decision maker, a confidence gap in the results can occur.

Visual Interactive Simulation

- Visual interactive simulation (VIS), visual interactive modeling (VIM) and visual interactive problem solving, is a simulation method that lets decision makers see what a model is doing, how it interacts with made decisions.
- Users employ knowledge to try different decision strategies while interacting with the model. Decision makers can contribute to model validation.
- VIS uses animated computer graphic displays to present the impact of different managerial decisions. It differs from regular graphics in that the user can adjust the decision-making process and see results of the intervention.
- VIS can represent static or dynamic systems. Static models display a visual image of the result of one decision alternative at a time. Dynamic models display evolving systems over time. The evolution is represented by animation.

Visual Interactive Models (VIM) and DSS

- **VIM in DSS** has been used in several operations management decisions.
- The method consists of priming a visual interactive model of a company with its current status.
- Waiting-line management is a good example of VIM. Such a DSS usually computes measures of performance for the various decision alternatives.
- Complex waiting-line problems require simulation. VIM can display the size of the waiting line as it changes during the simulation runs and can graphically present the answers to what-if questions regarding changes in input variables.
- The VIM approach can be used with AI. Integration of the two techniques adds several capabilities that range from the ability to build systems graphically to learning about the dynamics of the system.

Reference

- Ramesh Sharda, Dursun Delen, Efraim Turban; Analytics, Data Science, & Artificial Intelligence: Systems For Decision Support, 11ed, Global Edition, Pearson Education Limited, 2021, ISBN 10: 1 - 2 92-34155-6, ISBN 13: 978-1-292-34155-2

  Chapter 8 Prescriptive Analytics: Optimization and Simulation

Recommended Reading

- ***Artificial Intelligence: Systems for Decision Support": Monte Carlo Simulation***

https://www.palisade.com/risk/monte_carlo_simulation.asp

- ***It's Only Logical: Decision Tables and Decision Trees***

https://medium.com/analysts-corner/its-only-logical-decision-tables-and-decision-trees-12a8b52243ea

- **Decision Table: Clearer Logic and Better Decision**

  https://warren2lynch.medium.com/decision-table-clearer-logic-and-better-decision-603c3cff9f24

Recommended Video

- **Decision Trees & Decision Tables**

https://www.youtube.com/watch?v=A5-w3mof-3I

- **Monte Carlo Simulations in Excel**

  https://www.youtube.com/watch?v=UeGncSFijUM

- **Linear Programming (LP) Optimization with Excel Solver**

  https://www.youtube.com/watch?v=6xa1x_Iqjzg